C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
`http://c-immsim.iac.rm.cnr.it (alias to http://kraken.iac.rm.cnr.it/C-IMMSIM)`

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
tt doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
`https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262i`

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
`https://jitc.bmj.com/content/9/5/e002694`

Original C-IMMSIM model: `www.iac.cnr.it/∼filippo/c-immsim`

GETTING HELP:
Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)
Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

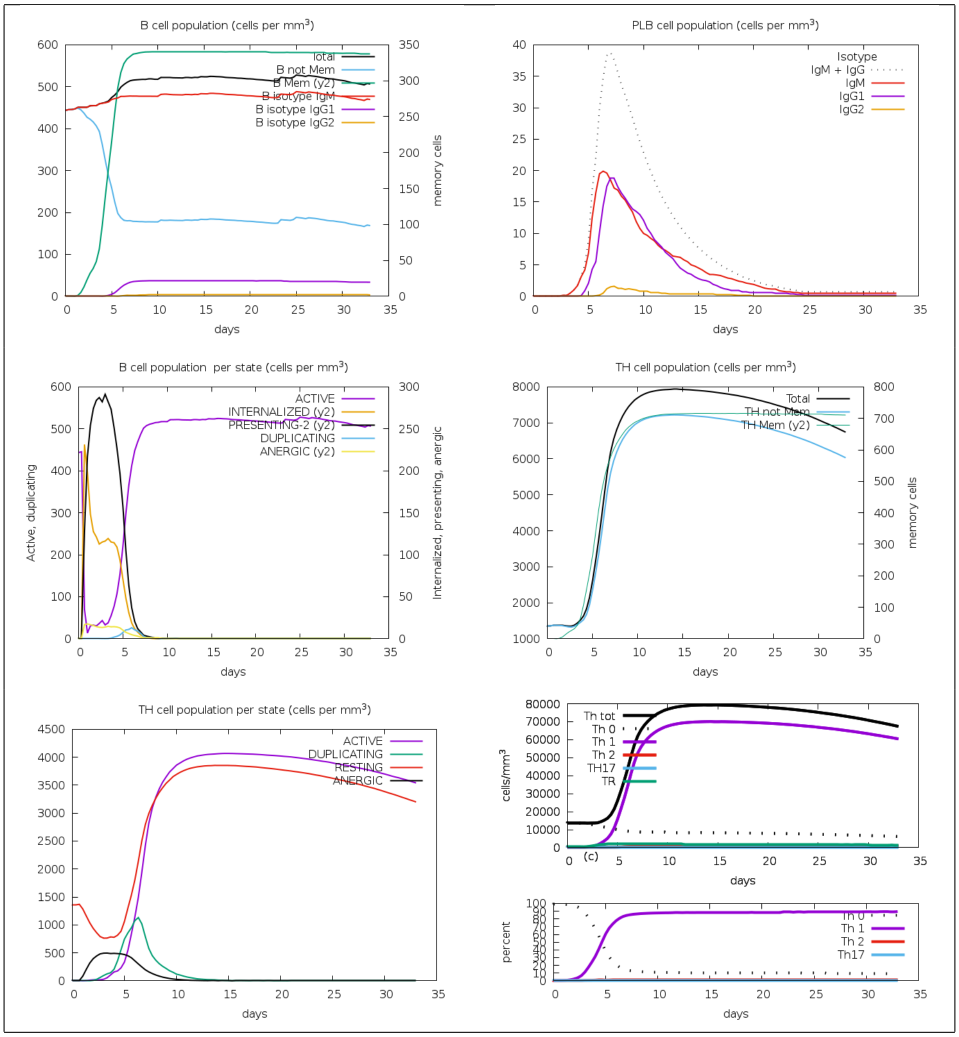

Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

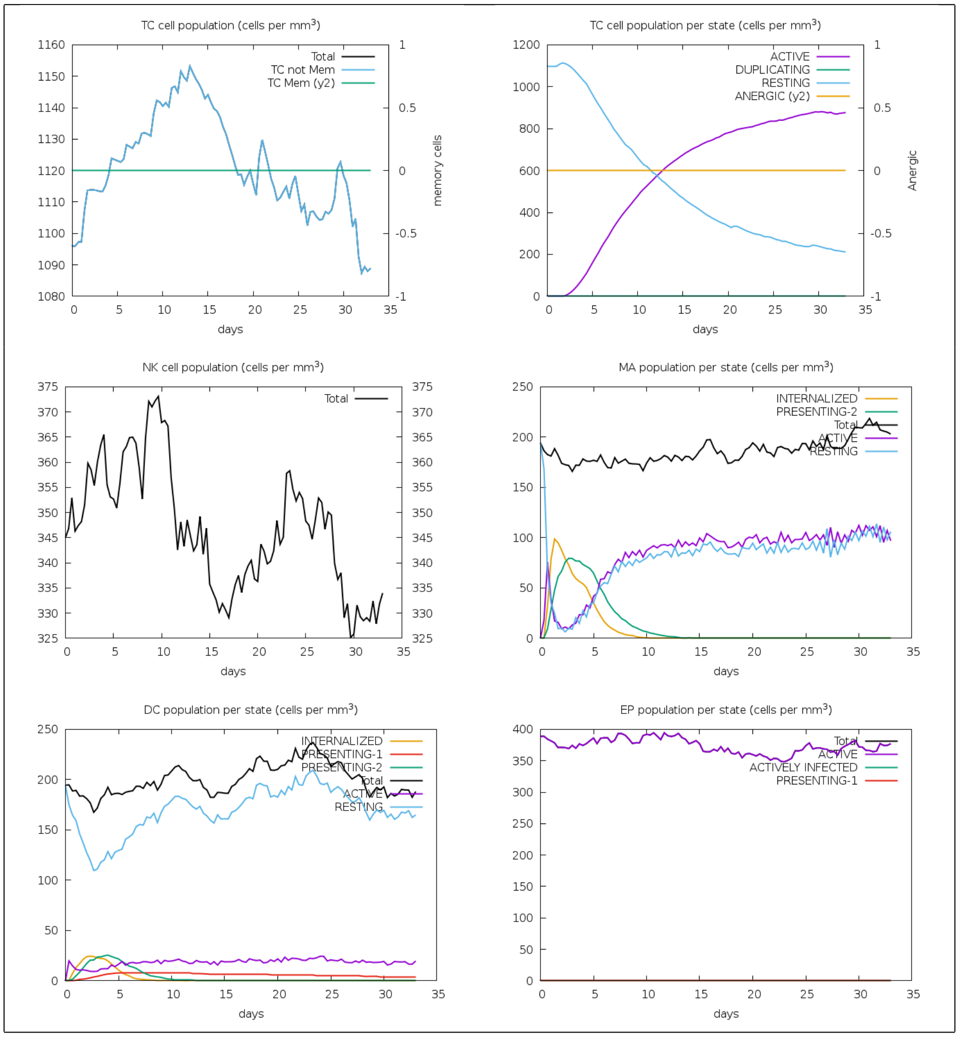

Figure 2: Legend: symbols as figure above.

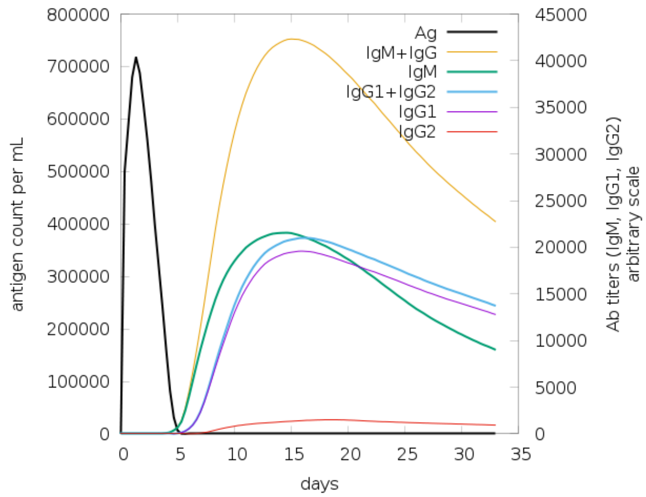

Figure 3: The virus, the immunoglobulins and the immunocomplexes.

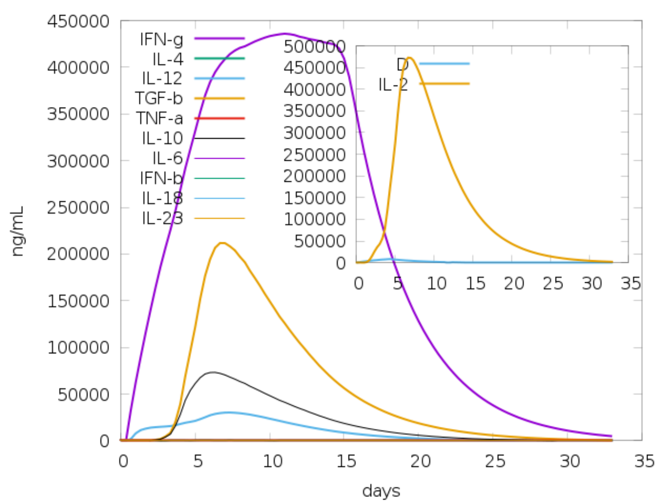

Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

```
Parker_B_scale_epitope_find
Use Parker's propensity scale, takes an antigen block as input,
and creates a list of residues that are possible epitopes.


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013202_5_NABpTu76u1Vx.FSA_1_001
----------------------------------


AKFVAAWTLKAAAGGSGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGA
GAEAAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGPGRHHLLVSGAEAAAKLPPWFPPMV

00000000000011111111110001111111111111111111111111111111111111111111111111111101
11100111000000011111111111111111111111111111111111111111111111111111111111111111
100000000000000000000000000000001100000000000011111110000000000000000000000000
Epitopes of protein 0 ------------
1]       pos=12 len=10        AGGSGPGTGP
2]       pos=25 len=53        LGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRH
3]       pos=79 len=4        DGVR
4]       pos=95 len=66        GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAG
5]       pos=207 len=7        TEKGPGP
```

```
DoPeptideList_I:
        Given the antigen injected creates the list of peptides for all the
        NumAgProts proteins and for all i.e., 4 MHCI molecules

        Read class I peptide list from file? NO

==========================================================================

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000


----------------------------------
Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013202_5_NABpTu76u1Vx.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGA
GAEAAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGPGRHHLLVSGAEAAAKLPPWFPPMV

Epitopes of protein 0 ------------
0]       pos=  -1 score=1.000000 unnormalised=104.8410000000        non-binding event

==========================================================================

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000


----------------------------------
```

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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013202_5_NABpTu76u1Vx.FSA_1_001
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AKFVAAWTLKAAAGGSGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGA
GAEAAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGPGRHHLLVSGAEAAAKLPPWFPPMV

Epitopes of protein 0 ------------
0]        pos=  -1 score=1.000000 unnormalised=104.8410000000        non-binding event

==========================================================================

Allele: B0702
Pseudo sequence: KAAREEQQIKAQTRE
Threshold: 8.702800
Max score: 28.406000


----------------------------------
Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013202_5_NABpTu76u1Vx.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGA
GAEAAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGPGRHHLLVSGAEAAAKLPPWFPPMV

Epitopes of protein 0 ------------
0]        pos=  64 score=0.002728 unnormalised=0.3012000000        RPGAPGGSG
1]        pos=  83 score=0.017497 unnormalised=1.9322000000        RPQKRPSCI
2]        pos= 190 score=0.004023 unnormalised=0.4442000000        GPGPGTSKF
3]        pos= 210 score=0.026354 unnormalised=2.9102000000        GPGPGRHHL
4]        pos=  -1 score=0.949399 unnormalised=104.8410000000        non-binding event

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Pseudo sequence: KAAREEQQIKAQTRE
Threshold: 8.702800
Max score: 28.406000


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013202_5_NABpTu76u1Vx.FSA_1_001
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AKFVAAWTLKAAAGGSGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGA
GAEAAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGPGRHHLLVSGAEAAAKLPPWFPPMV

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0]        pos=  64 score=0.002728 unnormalised=0.3012000000        RPGAPGGSG
1]        pos=  83 score=0.017497 unnormalised=1.9322000000        RPQKRPSCI
2]        pos= 190 score=0.004023 unnormalised=0.4442000000        GPGPGTSKF
3]        pos= 210 score=0.026354 unnormalised=2.9102000000        GPGPGRHHL
4]        pos=  -1 score=0.949399 unnormalised=104.8410000000        non-binding event
```

```
DoPeptideList_II:
        Given the antigen injected creates the list of peptides for all the
        NumAgProts proteins and for all i.e., 2 MHCII molecules
```

```
        Read class II peptide list from file? NO

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Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013202_5_NABpTu76u1Vx.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGA
GAEAAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGPGRHHLLVSGAEAAAKLPPWFPPMV

Epitopes of protein 0 ------------
0]        pos=   2 score=0.040812 unnormalised=3.3845600000        FVAAWTLKA
1]        pos=   6 score=0.005566 unnormalised=0.4615600000        WTLKAAAGG
2]        pos=   8 score=0.040800 unnormalised=3.3835600000        LKAAAGGSG
3]        pos= 173 score=0.077469 unnormalised=6.4245600000        YTLGPGPGT
4]        pos= 198 score=0.011366 unnormalised=0.9425600000        FLMGTYKRV
5]        pos= 218 score=0.026113 unnormalised=2.1655600000        LLVSGAEAA
6]        pos=  -1 score=0.797874 unnormalised=66.1680000000        non-binding event

========================================================================
```


```
Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000


----------------------------------
Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013202_5_NABpTu76u1Vx.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGA
GAEAAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGPGRHHLLVSGAEAAAKLPPWFPPMV

Epitopes of protein 0 ------------
0]        pos=   2 score=0.040812 unnormalised=3.3845600000        FVAAWTLKA
1]        pos=   6 score=0.005566 unnormalised=0.4615600000        WTLKAAAGG
2]        pos=   8 score=0.040800 unnormalised=3.3835600000        LKAAAGGSG
3]        pos= 173 score=0.077469 unnormalised=6.4245600000        YTLGPGPGT
4]        pos= 198 score=0.011366 unnormalised=0.9425600000        FLMGTYKRV
5]        pos= 218 score=0.026113 unnormalised=2.1655600000        LLVSGAEAA
6]        pos=  -1 score=0.797874 unnormalised=66.1680000000        non-binding event
```
